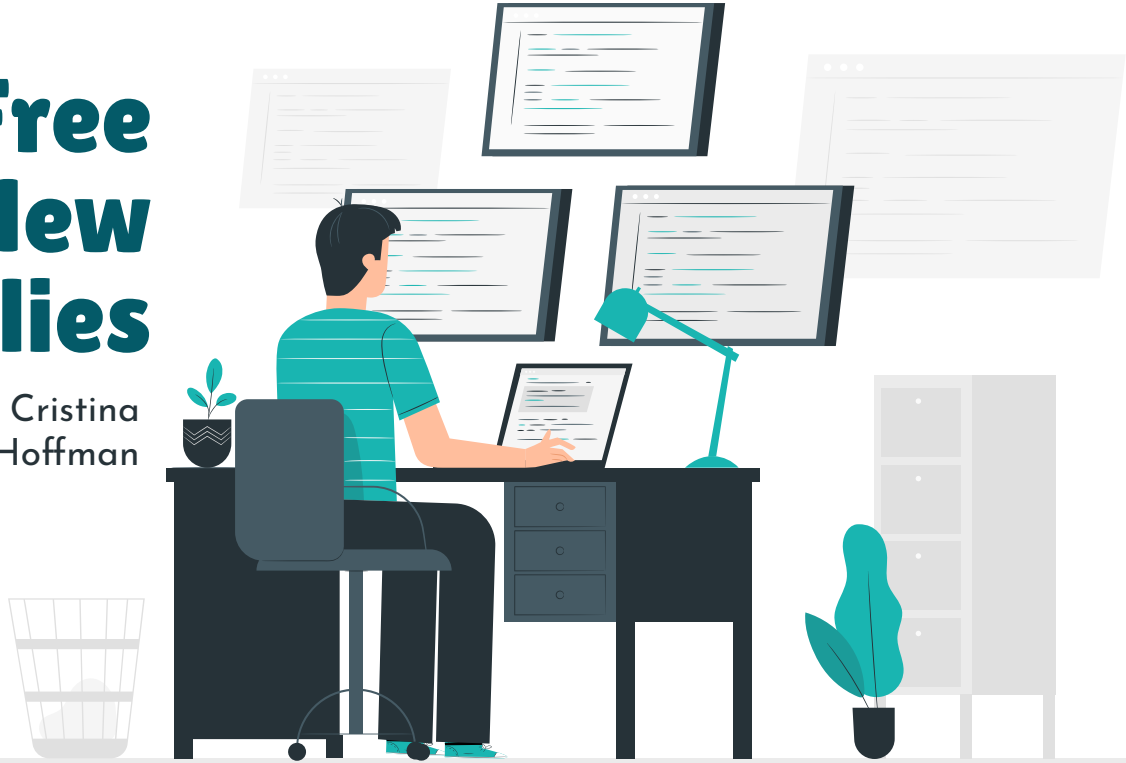
The Effect of Free Child Care on New Mexico Families

Ava Pearson, Catherine Devens, Cristina Jackson, Cy Coldiron, and Octavia Hoffman

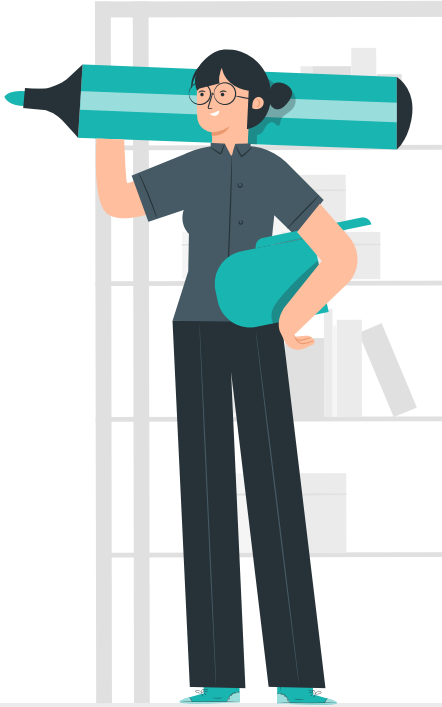


Motivation

- Many moms must choose child care over career goals.
- Parents with kids under 12 often face a choice: pay high costs or leave work.
- 1.2–1.5M moms cut hours monthly due to child care (KPMG).
- Without child care, parents lose income or leave their jobs.
- This is not just a family issue—it hurts the economy.
  a. U.S. businesses lose $12.7B yearly from child care gaps.
  b. The total economic loss is around $57B each year.

Motivation Cont'd

- Sweden treats child care as a public service, not a personal issue.
- All kids get affordable, high-quality care from age one, with fees tied to income.
- Paid leave and universal preschool let parents work without hurting family stability.
- The result: high maternal employment, less stress, and better early learning.
- The U.S. lacks a universal system, but CCAP helps fill the gap.
- CCAP supports low-income families, helping parents stay employed and boost the economy.



Research Question

What is the impact of New Mexico's free childcare policy on the labor market behavior of qualifying mothers?

First Order Effect

How did these policies impact the number of working hours for qualifying mothers?

Review of Existing Literature

Study #1

Authors: Thomas Cornelissen, Christian Dustmann, Anna Raute, and Uta Schönberg
Year: 2018

- Policy: 1990s German Child Care Reform
- Natural experiment: staggered expansion of subsidized child care.
- Findings: disadvantaged children have large gains in school readiness.

Study #2

Authors: Pierre Lefebvre, Philip Merrigan, and Matthieu Verstraete
Year: 2009

- Policy: 1997 Québec Universal Low-Fee Childcare
- Natural experiment: Use other Canadian Provinces as controls.
- Findings: Long-term labor supply gains for mothers, especially for low-income families.

Literature Review Continued...

Effects of Subsidized Child Care Vary by Context:

- **Maternal Employment**:
  - Québec: Increased employment.
  - Finland: Slight decrease with home care subsidies.
  - Norway: Little effect.
- **Child Outcomes**:
  - Norway: Improved academic performance.
  - Québec: Some negative behavioral effects.

Understanding the impact of New Mexico's policy on family outcomes will help assess child care subsidies in the U.S. context.

Explanation of Data Sources

- **Data Source**:
  - ***American Community Survey (ACS)*** via **IPUMS** USA, (Ruggles et a., 2025).
    - Collects data monthly and releases it annually from a rotating 1% sample of U.S population
- **Why ACS?**:
  - More up-to-date income and employment information than other federal surveys (which often lag by a year or more)
- **Observation Unit**:
  - Each row is an individual, enabling us to isolate mothers with children under 12 for treatment and control comparisons.

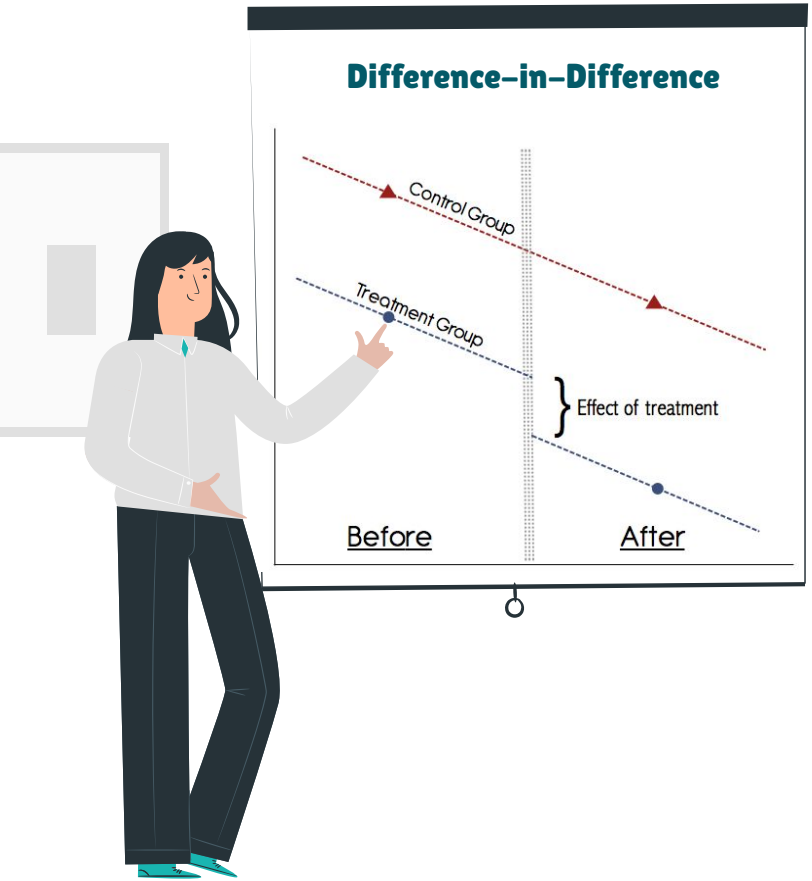
Explanation of Data Sources

In our data, we filtered for...

- Sex: Female
- Employment Status: Employed
- Age of youngest child: 12 or under
- Location: New Mexico
- Time: 2021 -2023

We then evaluated for...

- Number of observations
- Hours per week spent working
- Family Income
- Wage Income
- Age



Methodology/Empirical Strategy

Strategy: Difference-in-Differences (DiD) comparing changes in average hours worked

Shock = The May 1, 2022 Policy Change

Pre-treatment: **2021 & 2022**
Post-Treatment: **2023**

Control Group = Mothers **ineligible** for the Childcare Assistance Program with Income greater than 400% of the Federal Poverty Level.

Treatment Group = Mothers **eligible** for the Childcare Assistance Program with Income less than or equal to 400% of the Federal Poverty Level.

All families enrolled in the state's Child Care Assistance Program will no longer owe copays for child care services, making child care cost-free.

Summary Statistics

Summary Statistics by Treatment Status and Period (2021–2023)

Mothers with Children Under 12 in New Mexico

Period	Observations	Mean Weekly Hours	SD Weekly Hours	Mean Family Income	SD Family Income	Mean Wage Income	SD Wage Income	Mean Age	SD Age
Control (> 400% FPL)									
Post (2023)	445	30.3	17.9	188,281	89,519	51,158	55,572	42.6	12.0
Pre (2021–2022)	828	31.8	18.1	195,062	110,654	59,173	64,736	42.1	12.0
Treated (≤ 400% FPL)									
Post (2023)	1059	22.3	19.0	54,133	30,232	18,755	36,913	39.2	13.0
Pre (2021–2022)	2132	21.0	19.5	50,760	30,528	16,541	29,756	39.1	12.6

Income values are reported in nominal dollars from the prior calendar year (e.g., 2022 ACS = 2021 income).

Treatment group = family income ≤ 400% of the Federal Poverty Level (FPL); Control group = family income > 400% FPL.

Regression Specification

$$Y_{it} = \beta_0 + \beta_1 TREAT_i + \beta_2 POST_t + \beta_3 (TREAT_i \times POST_t) + \varepsilon_{it}$$

Coefficient	Variable	Interpretation
β_0	Control Group (2021–2022)	Mean weekly hours for the control group in the pre-policy period.
β_1	Treated vs. Control (Pre)	Difference in mean weekly hours between treated and control groups before the policy.
β_2	Control Group Change (2023)	Change in control group's weekly hours from pre to post.
β_3	Treatment Effect (DiD Estimate)	DiD estimate: additional effect for treated group in 2023 relative to control.

Note: i indexes individuals (mothers with children under 12 in New Mexico).

Difference–in–Difference Regression Results

- **Pre-Policy (2021-2022):**
  - Control worked **31.78 hours/week**
  - Treated mothers worked **10.81 fewer hrs/week**
- **Post-Policy (2023)**
  - Control group hours **declined 1.48 hrs**
  - Treated mothers **increased weekly hours by 2.85** (statistically significant)

Difference-in-Differences Regression (2021–2023)

Weekly Hours Worked, Mothers with Children Under 12 in New Mexico

Variable	Estimate	Std. Error	p-value
Treatment Effect (DiD Estimate)	2.85*	1.28	0.026
Control Group (2021–2022)	31.78***	0.63	0.000
Treated vs. Control (Pre)	-10.81***	0.76	0.000
Control Group Change (2023)	-1.48	1.06	0.161

*Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$*

Visualization

Pre-policy trends (2021-2022):

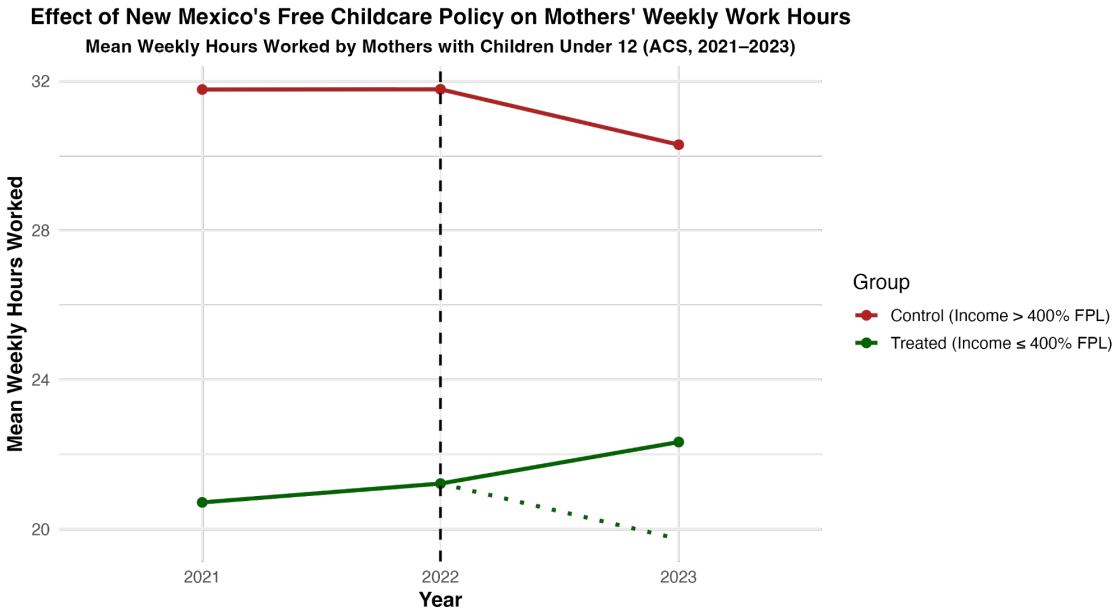
- **Parallel**, supporting the validity of the DiD approach.

Closing gap in Post-Policy Period:

- Difference in hours between two groups narrows (effective policy?)

Counterfactual Line:

- Projected work hours for treated group had they followed the same trend as the control



Treated = mothers with income ≤ 400% FPL; Control = income > 400% FPL. Policy enacted in 2022. Dotted line is counterfactual projection.

Conclusion

Possible Second Order Effects:

- Income
- Educational performance
- **Economic participation**
- Job retention
- Mental health implications
- Crime and juvenile detention

Limitations:

- **Covid-19** recovery
- Stimulus packages
- GDP increase by 5.4%
- Different people in **pre & post periods**
- Inflation

Conclusion & Potential Next Steps

- **Using More Data as It Becomes Available:**
  - Track future employment rates to see if the pattern persists.
  - Collect data on second-order effects as they become measurable with time.

- **Checking for Robustness:**
  - Compare results using different control groups (e.g., states with similar demographics) to test if trends hold.

- **Cost-Benefit Analysis:**
  - Determine if gains from maternal employment outweigh childcare subsidy costs.

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